

Assignment - 5

1. Print numbers from 1 to 5 using a for loop.

```
#include <stdio.h>
int main()
{
    int i;
    for (i = 1; i <= 5; i++)
    {
        printf("%d", i);
    }
    return 0;
}
```

Output

1 2 3 4 5

2. Print numbers from 1 to 5 using a while loop.

```
#include <stdio.h>
int main()
{
    int i = 1;
    while (i <= 5)
    {
        printf("%d\n", i);
        i++;
    }
    return 0;
}
```

Output

1
2
3
4
5

3. Print numbers from 1 to 5 using a do while loop.

```
#include <stdio.h>
int main()
{
    int i = 1;
    do {
        printf("%d", i);
        i++;
    } while (i <= 5);
    return 0;
}
```

output

1 2 3 4 5

4. Print even numbers from 2 to 10 (using for loop, using while loop, using do...while loop).

for loop

```
#include <stdio.h>
int main()
{
    int i;
    for (i=1; i<=10; i++)
    {
        if (i%2==0)
            printf("%d", i);
    }
    return 0;
}
```

Output :- 2 4 6 8 10

while loop

```
#include <stdio.h>
int main()
{
    int i = 1;
    while (i <= 10) {
        if (i%2==0)
            printf("%d", i);
        i++;
    }
    return 0;
}
```

Output :- 2 4 6 8 10

do...while loop

```
#include <stdio.h>
int main()
{
    int i = 1;
    do {
        if (i%2==0)
            printf("%d", i);
    } while (i <= 10);
    return 0;
}
```

Output :- 2 4 6 8 10

5. Print the sum of numbers from 1 to 5 using while, for, do-while).

```
#include <stdio.h>
int main()
{
    int i, m = 0;
    for (i = 1; i <= 5; i++)
    {
        m += i;
    }
    printf("Sum of 1 to 5 is : %d", m);
    return 0;
}
```

Output :- Sum of 1 to 5 is : 15

while

```
#include <stdio.h>
int main()
{
    int i = 1, m = 0;
    while (i <= 5)
    {
        m = m + i;
        i = i + 1;
    }
    printf("Sum of 1 to 5 is : %d", m);
    return 0;
}
```

Output :- Sum of 1 to 5 is : 15

do while

```
#include <stdio.h>
int main()
{
    int i = 1, m = 0;
    do {
        m += i;
        i++;
    } while (i <= 5);
    printf("Sum of 1 to 5 : %d", m);
    return 0;
}
```

Output :- sum of 1 to 5 : 15

6. Print the multiplication table of 3 (from 3×1 to 3×10) (using for loop, while, do while).

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int i=0, m;
```

```
for (i=1; i<=10; i++)
```

```
{
```

```
m = 3 * i;
```

```
printf ("3 x %d = %d", i, m);
```

```
}
```

```
return 0;
```

```
}
```

Output

$3 \times 1 = 3$

$3 \times 2 = 6$

$3 \times 3 = 9$

$3 \times 4 = 12$

$3 \times 5 = 15$

$3 \times 6 = 18$

$3 \times 7 = 21$

$3 \times 8 = 24$

$3 \times 9 = 27$

$3 \times 10 = 30$

While

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int i=1, m;
```

```
while (i<=10)
```

```
{
```

```
m = 3 * i;
```

```
printf ("3 x %d = %d", i, m);
```

```
i = i+1;
```

```
}
```

```
return 0;
```

```
}
```

Output:-

$$\begin{array}{l} 3 \times 1 = 3 \\ 3 \times 2 = 6 \\ 3 \times 3 = 9 \\ 3 \times 4 = 12 \\ 3 \times 5 = 15 \\ 3 \times 6 = 18 \\ 3 \times 7 = 21 \\ 3 \times 8 = 24 \\ 3 \times 9 = 27 \\ 3 \times 10 = 30 \end{array}$$

do while

```
#include <stdio.h>
int main()
{
    int i=1, m;
    do { m = 3 * i;
        printf("3 x %d = %d \n", i, m);
        i = i + 1;
    } while (i <= 10);
    return 0;
}
```

Output:-

$$\begin{array}{l} 3 \times 1 = 3 \\ 3 \times 2 = 6 \\ 3 \times 3 = 9 \\ 3 \times 4 = 12 \\ 3 \times 5 = 15 \\ 3 \times 6 = 18 \\ 3 \times 7 = 21 \\ 3 \times 8 = 24 \\ 3 \times 9 = 27 \\ 3 \times 10 = 30 \end{array}$$

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